Chandravardhan Singh Raghaw

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AI Centre of Excellence in Healthcare, Indian Institute of Science, Bengaluru, India

Research Profile

My research focuses on multimodal AI for medical imaging, with an emphasis on vision-language modeling and report generation. I develop contrastive and knowledge-guided methods to enable reliable, interpretable, and scalable clinical decision support systems.

Keywords: Vision-Language Modeling | Medical Imaging | Multimodal Learning | Report Generation.

Education

- **Indian Institute of Technology Indore** *2022 - 2026*
  PhD - Computer Science and Engineering CGPA: 9.00 CPI
  - *Thesis:* "Advancing Computer-Aided Medical Diagnosis: Synthesizing Global Context and Local Precision for Pathological Image Analysis of Diverse Imaging Modalities" (Defended: August 2026).

- **Govt. Engineering College, Ajmer (RTU Kota)** *2011-2013*
  M.Tech - Information Technology (Topper in Department) Percentage: 70%

- **Govt. Engineering College, Ajmer (RTU Kota)** *2007 - 2011*
  B.Tech - Information Technology Percentage: 65%

- **Lawrence and Mayo Public School, Ajmer (CBSE)** *2006 - 2007*
  Higher Secondary Education - Class XII Percentage: 66.80%

- **Lawrence and Mayo Public School, Ajmer (CBSE)** *2004 - 2005*
  Secondary Education - Class X Percentage: 72.20%

Experience

- **AI Centre of Excellence in Healthcare (TANUH), IISc: Postdoctoral Researcher** *Jul 2026 – Present*
  Focus: Multimodal AI, Foundation Models, Medical Vision-Language
  - Researching Indian brain foundation models (IndiNeuroFM) for clinical healthcare at TANUH.
  - Developing multimodal vision-language architectures for automated clinical report generation.
  - Designing robust evaluation benchmarks and cross-modal alignment frameworks for medical imaging.

- **Robo India: Web Developer** *2018 – 2022*
  Tools: PHP, JavaScript, MySQL, HTML5/CSS3
  - Developed responsive web applications and e-commerce solutions tailored to client specifications.
  - Managed full SDLC, including UI/UX design, database integration, and performance optimization.
  - Automated maintenance tasks and provided technical support, ensuring maximum platform uptime.

- **Govt. Engineering College Ajmer: Assistant Professor (On-contract)** *2011 – 2013, 2015 – 2018*
  Focus: Software Engineering, Algorithms, Operating Systems
  - Delivered B.Tech and MCA lectures in core CS subjects, improving student academic performance.
  - Orchestrated departmental administration, curriculum development, and examination coordination.
  - Mentored students in final year projects and modernized laboratory sessions with UML modeling.

- **Mechatronix Techno Solution: Project Manager (IT)** *2014 – 2015*
  Tools: MS Project, Agile Methodologies, Resource Allocation
  - Spearheaded IT infrastructure projects, ensuring delivery within strict timelines and budgets.
  - Optimized resource allocation and vendor contracts to reduce operational overhead.
  - Liaised between stakeholders and technical teams to align project milestones and goals.

- **Vivekananda Institute of Technology Jaipur: Assistant Professor** *2013 – 2014*
  Focus: Undergraduate courses – Computer Architecture, Operating Systems
  - Taught undergraduate courses in Computer Architecture and Advanced Topics in Operating Systems.
  - Advised academic group on career development and professional IT certifications.
  - Evaluated student technical progress through continuous assessment and constructive feedback.

List of Publications

▶ **Conference**

1. **Chandravardhan Singh Raghaw**, Shahid Shafi Dar, Arnav Nishith Sharma, Shrey Modi, and Nagendra Kumar. "CliQ-RRG: Clinical-Knowledge Guided Disease-aware Visual–Textual Alignment for QA-Style Radiology Report Generation." **Findings of the Association for Computational Linguistics: EMNLP 2026**, pp. 50153–50171, Oct. 2026, doi: 10.18653/2026.findings-emnlp.2333. *[CORE Rank: A*].*

▶ **Journals**

2. **Chandravardhan Singh Raghaw**, Anushka Parwal, Shahid Shafi Dar, Prajakta Darade, Nagendra Kumar. "H-SemiS: Hierarchical Fusion of Semi and Self-Supervised Learning for Knee Osteoarthritis Severity Grading." **Expert Systems with Applications**, vol. 322, p. 132279, Aug. 2026, doi: 10.1016/j.eswa.2026.132279. *[Impact Factor: 7.5 (2026), Q1 (SJR 2026)].*

3. **Chandravardhan Singh Raghaw**, Jasmer Singh Sanjotra, Mohammad Zia Ur Rehman, Shubhi Bansal, Shahid Shafi Dar, and Nagendra Kumar, "T-MPEDNet: Unveiling the Synergy of Transformer-aware Multiscale Progressive Encoder-Decoder Network with Feature Recalibration for Tumor and Liver Segmentation," **Biomedical Signal Processing and Control**, vol. 110, p. 108225, Dec. 2025, doi: 10.1016/j.bspc.2025.108225. *[Impact Factor: 4.9 (2025), Q1 (SJR 2025)]*

4. **Chandravardhan Singh Raghaw**, Aryan Yadav, Jasmer Singh Sanjotra, Shalini Dangi, and Nagendra Kumar. "MNet-SAt: A multiscale network with spatial-enhanced attention for segmentation of polyps in colonoscopy." **Biomedical Signal Processing and Control**, vol. 102, p. 107363, Apr. 2025, doi: 10.1016/j.bspc.2024.107363. *[Impact Factor: 4.9 (2025), Q1 (SJR 2025)].*

5. **Chandravardhan Singh Raghaw**, Parth Shirish Bhore, Mohammad Zia Ur Rehman, and Nagendra Kumar. "An explainable contrastive-based dilated convolutional network with transformer for pediatric pneumonia detection.", **Applied Soft Computing**, vol. 167, p. 112258, Dec. 2024, doi: 10.1016/j.asoc.2024.112258. *[Impact Factor: 7.2 (2024), Q1 (SJR 2024)].*

6. **Chandravardhan Singh Raghaw**, Arnav Sharma, Shubhi Bansal, Mohammad Zia Ur Rehman, and Nagendra Kumar. "CoTCoNet: An optimized coupled transformer-convolutional network with an adaptive graph reconstruction for leukemia detection." **Computers in Biology and Medicine**, vol. 179, p. 108821, Sep. 2024, doi: 10.1016/j.compbiomed.2024.108821. *[Impact Factor: 7.7 (2024), Q1 (SJR 2024)].*

7. **Chandravardhan Singh Raghaw**, Tanisha Jitendra Sahu, Prajakta Darade, Shahid Shafi Dar, and Nagendra Kumar. "Quantum-Guided Textural–Spatial Fusion Network for Thyroid Nodule Classification in Ultrasound Imaging." **International Journal of Imaging Systems and Technology**, vol. 36, no. 5, p. e70428, 2026, doi: 10.1002/ima.70428. *[Impact Factor: 3.3 (2024), Q2 (SJR 2024)].*

8. Shalini Dangi, Surya Karthikeya Mullapudi, **Chandravardhan Singh Raghaw**, Shahid Shafi Dar, Mohammad Zia Ur Rehman, and Nagendra Kumar. "A multi-temporal multi-spectral attention-augmented deep convolution neural network with contrastive learning for crop yield prediction." **Computers and Electronics in Agriculture**, vol. 239, p. 110895, Dec. 2025, doi: 10.1016/j.compag.2025.110895. *[Impact Factor: 8.9 (2025), Q1 (SJR 2025)].*

9. Shahid Shafi Dar, Bharat Kaurav, Arnav Jain, **Chandravardhan Singh Raghaw**, Mohammad Zia Ur Rehman, and Nagendra Kumar. "An explainable deep neural network with frequency-aware channel and spatial refinement for flood prediction in sustainable cities." **Sustainable Cities and Society**, vol. 130, p. 106480, Jul. 2025, doi: 10.1016/j.scs.2025.106480. *[Impact Factor: 12.0 (2025), Q1 (SJR 2025)].*

10. Mohammad Zia Ur Rehman, Devraj Raghuvanshi, Harshit Pachar, **Chandravardhan Singh Raghaw**, and Nagendra Kumar. "Hierarchical attention-enhanced contextual CapsuleNet for multilingual hope speech detection." **Expert Systems with Applications**, vol. 268, p. 126285, Apr. 2025, doi: 10.1016/j.eswa.2024.126285. *[Impact Factor: 7.5 (2025), Q1 (SJR 2025)].*

11. Shubhi Bansal, Mohit Kumar, **Chandravardhan Singh Raghaw**, and Nagendra Kumar. "Sentiment and hashtag-aware attentive deep neural network for multimodal post popularity prediction." **Neural Computing and Applications**, vol. 37, no. 4, pp. 2799–2824, Dec. 2024, doi: 10.1007/s00521-024-10755-5. *[Impact Factor: 4.5 (2025), Q1 (SJR 2025)].*

12. Shubhi Bansal, Kushaan Gowda, Mohammad Zia Ur Rehman, **Chandravardhan Singh Raghaw**, and Nagendra Kumar. "A hybrid filtering for micro-video hashtag recommendation using graph-based deep neural network." **Engineering Applications of Artificial Intelligence**, vol. 138, p. 109417, Dec. 2024, doi: 10.1016/j.engappai.2024.109417. *[Impact Factor: 8.0 (2026), Q1 (SJR 2026)].*

▶ **Under Review: Conference**

13. **Chandravardhan Singh Raghaw**, Manan Vaishnav, Saurabh Sharma, Shahid Shafi Dar, and Pramod Singh Rathore. "UTTAR: Query-Guided Ultrasound Visual–Textual Alignment with Structured Reasoning for QA-Style Report Generation." *Under review* at NAACL (2027). [Core Rank: A]

14. **Chandravardhan Singh Raghaw**, Manan Vaishnav, Saurabh Sharma, and Pramod Singh Rathore. "Disease-Aware Multimodal Alignment with Clinical Knowledge Retrieval for Chest X-Ray Report Generation." *Under review* at National Conference on Communications (2027). [National Conference]

15. Shahid Shafi Dar, Mihir Kanchan Karandikar, **Chandravardhan Singh Raghaw**, Pranjal Pandey, and Nagendra Kumar. "Grounded, Not Learned: Corpus-Induced Topic Anchors and Non-Monotonic Reliability Gating for Multimodal Crisis Informatics." *Under review* at ACM TheWebConf (2026). [Core Rank: A*]

16. Shahid Shafi Dar, Dhanvin Sanjay Namboodiri, **Chandravardhan Singh Raghaw**, and Nagendra Kumar. "Terrain-Guided Graph Refinement with Kernelized Global–Local Interaction for Semi-Supervised Flood Segmentation." *Under review* at AAAI (2027). [Core Rank: A*]

► **Under Review: Journal**

17. **Chandravardhan Singh Raghaw**, Manan Vaishnav, and Phaneendra K. Yalavarthy. "RDV: Report, Detail, Verify – Evidence-Grounded Brain CT Report Generation with a Self-Calibrating Multimodal Large Language Model." Under review at *Medical Image Analysis* (2027).

18. **Chandravardhan Singh Raghaw**, Gaurav Mahendraker, G Aakash, Shahid Shafi Dar, and Nagendra Kumar. "Consistency-Regularized Causal Teacher–Student Learning with Diffusion Synthesis for PCOS Diagnosis." Under review at *IEEE Transactions on Emerging Topics in Computational Intelligence* (2026).

19. **Chandravardhan Singh Raghaw**, Shahid Shafi Dar, Mohammad Zia Ur Rehman, Neerupam, Sanjeet Kumar, Eeshan Bablani, and Nagendra Kumar. "CXR4MTL: A Multi-Task Chest X-Ray Dataset and Framework for Classification, Segmentation, Report Generation, and Answer Extraction." Under review at *IEEE Journal of Biomedical and Health Informatics* (2026).

20. **Chandravardhan Singh Raghaw**, Naman V. Shetty, Karanam Venkata Lakshmi Sarath Chandra, Divyae Arya, Shahid Shafi Dar, and Nagendra Kumar. "AbDo-CTRG: A Dataset and Baseline for Abdominal CT Report Generation." Under review at *ACM Transactions on Computing for Healthcare* (2026).

21. **Chandravardhan Singh Raghaw**, Parth Shirish Bhore, Shahid Shafi Dar, and Nagendra Kumar. "ClinIC-CTRG: Contrastive Alignment with Knowledge-augmentation and In-Context Learning for CT Report Generation." Under review at *Machine Vision and Applications* (2026).

22. **Chandravardhan Singh Raghaw**, Arnav Nishit Sharma, Shahid Shafi Dar, and Nagendra Kumar. "MCMD-Mamba: Multi-Channel Feature Enhancement with Multi-Direction Vision Mamba for Retinal Disease Classification." Under review at *Image and Vision Computing* (2026).

23. **Chandravardhan Singh Raghaw**, Shreeharish A, Manikandan S, Shahid Shafi Dar, Mohammad Zia Ur Rehman, and Nagendra Kumar. "From Early CNNs to Recent Transformers and LLMs for Pneumonia Detection: A Comprehensive Survey of Taxonomy, Methods, Datasets, Challenges and Future Trends." Under review at *Journal of Imaging Informatics in Medicine* (2026).

24. Manav Vaishnav*, **Chandravardhan Singh Raghaw*** and Phaneendra K. Yalavarthy. "SAGE-CT: A Lightweight Retrieval-Augmented Framework with Slice-Aware Latent Compression for Head Computed Tomography Report Generation." Under review at *IEEE Journal of Biomedical and Health Informatics* (2027).

25. Saurabh Sharma, **Chandravardhan Singh Raghaw**, Priyanka Patil, and Phaneendra K. Yalavarthy. "Toward Sovereign Indian Brain Foundation Models: Opportunities, Challenges, and Future Directions." Under review at *Neurocomputing* (2027).

26. Shahid Shafi Dar, Arnav Jain, **Chandravardhan Singh Raghaw**, and Nagendra Kumar. "A Benchmark Dataset and Graph-Based Multi-Representation Learning with Tri-Axial Attention for Disaster Damage Assessment." Under review at *Information Fusion* (2026).

27. Bhavesh Agrawal, Samyak Dhyani, **Chandravardhan Singh Raghaw**, Shalini Dangi, Shahid Shafi Dar, Mohammad Zia Ur Rehman, and Nagendra Kumar. "M2D-MambaNet: Learning Local and Global Semantics with Multi-Scale Multi-Directional Mamba for Efficient Gastrointestinal Disease Detection." Under review at *Expert Systems With Applications* (2026).

► **Book Chapter**

28. **Chandravardhan Singh Raghaw**, Shahid Shafi Dar, Mohammad Zia Ur Rehman, and Nagendra Kumar. "Explainable AI in Medical Imaging: A Comparative Study of Model-Agnostic Methods and Gradient-Based Class Activation Mapping Techniques for Classification and Segmentation." In ***Explainable AI for Transparent and Trustworthy Medical Decision Support***, pp. 311–330, Elsevier, 2027.

► **Preprints**

29. Shubhi Bansal, Sreekanth Madisetty, Mohammad Zia Ur Rehman, **Chandravardhan Singh Raghaw**, Gaurav Duggal, Nagendra Kumar. "A Comprehensive Survey of Mamba Architectures for Medical Image Analysis." arXiv preprint (2024).

30. Prasad Chaudhari, Aman Kumar, **Chandravardhan Singh Raghaw**, Mohammad Zia Ur Rehman, and Nagendra Kumar. "GCM-Net: Graph-enhanced Cross-Modal Infusion with a Metaheuristic-Driven Network for Video Sentiment and Emotion Analysis." arXiv preprint (2024).

Teaching Experience

- **Indian Institute of Technology Indore (Teaching Assistant)** *2022 – 2026*
  - IC 151: Computer Programming Lab — Autumn 2022, Spring 2023
  - CS 207: Database & Information Systems — Autumn 2023, Autumn 2024, Autumn 2025
  - CS 358: Compiler Techniques Lab — Spring 2024, Spring 2025
- **Government Engineering College, Ajmer** *2012 – 2018*
  - 6CS6.3A: Human Computer Interface — Spring 2012
  - 8CS1A: Mobile Computing — Spring 2013
  - M-303: Computer Graphics — Autumn 2015
  - M-401: Software Engineering & UML — Spring 2016, Spring 2017, Spring 2018
  - M-502: Analysis and Design of Algorithms — Autumn 2016
  - M-103: Operating Systems — Autumn 2017
- **Vivekananda Institute of Technology (ViT), Jaipur** *2013 – 2014*
  - 5CS1A: Computer Architecture — Autumn 2013
  - 6CS6.1A: Advanced Topics in Operating Systems — Spring 2014

Technical & Research Proficiencies

- **Programming Languages:** Python, C
- **Frameworks & Libraries:** PyTorch, TensorFlow, OpenCV, Scikit-learn
- **AI Specializations:** Vision-Language Models (VLMs), Large Language Models (LLMs), Generative AI (Diffusion, GANs), State Space Models (Mamba)
- **Research Methodologies:** Causal Inference, Contrastive Learning, Semi-Supervised Learning, Explainable AI (XAI), Image Segmentation & Classification
- **Tools & Systems:** LaTeX, CorelDRAW, SSH (PuTTY/WinSCP), Supervisely, Hugging Face

Student Mentorship

- Supervised 10+ students on projects in medical image analysis, vision-language modeling, and multimodal learning.
- Guided complete research workflows: problem formulation, modeling, experimentation, and writing.
- Supported students in developing reproducible deep learning systems using PyTorch and medical imaging datasets.

Academic Service

- **Reviewer (Conference):** MICCAI, ACM Web Conference.
- **Reviewer (Journal):** Expert Systems with Applications, Engineering Applications of Artificial Intelligence, Biomedical Signal Processing and Control, Informatics in Medicine Unlocked, Scientific Reports, BMC Artificial Intelligence, BMC Cancer BMC Medical Informatics and Decision Making, Multimedia Systems, Signal, Image and Video Processing, Discover Applied Sciences.

Honors and Awards

- **Best Research Presentation**: Symposium 4.0, Dept. of CSE, IIT Indore — *February 2026*
- **Best Poster Presentation**: Symposium 4.0, Dept. of CSE, IIT Indore — *February 2026*
- **Best Poster Presentation**: Research and Industrial Conclave (RIC 2026), IIT Indore — *January 2026*
- **Best Presentation Award**: IEEE CIS Summer School, IIT Indore — *July 2025*

Positions of Responsibility

- **Coordinator, Symposium 3.0**: Dept. of CSE, IIT Indore — *March 2025*
- **Coordinator, Research Symposium 1.0 2022-23**: Dept. of CSE, IIT Indore — *January 2023*
- **Coordinator, One Week Short Term Course**: Ubiquitous Computing, GEC Ajmer — *September 2016*
- **Organizing Committee Member**: National Workshop on Telemedicine, GEC Ajmer — *March 2013*

Certifications

- **UGC-NET 2024** : Qualified for Ph.D — *Scorecard*
- **GATE 2022** : Qualified with Rank - 6121) — *Scorecard*
- **Neural Networks and Deep Learning** : Coursera — *Link*
- **Machine Learning with Python** : freeCodeCamp — *Link*
- **NCC 'C' Certificate (Army Wing)** : 'A' Grade, Ministry of Defence, Govt. of India — *Link*
- **Other Certifications** — *View All*

Professional Development & Engagement

- **Research Presentation**: Oral Presentation in RIC, IIT Indore — *January 2026*
- **Winter School on Generative AI**: IEEE CIS, IIT Indore — *July 2025*
- **Advanced Signal Processing for Smart Health**: Workshop, IIT Indore — *November 2023*
- **Teaching Assistant Training Program**: Dept. of CSE, IIT Indore — *June 2023*
- **Art of Scientific Writing for PhD Students**: Workshop, IIT Indore — *September 2023*
- **Mental Healthcare and Artificial Intelligence**: Center for Depression Diagnosis, IIT Guwahati — *April 2022*
- **Supplementary Certifications**

References

Contact details available upon request.